Sources of variability in the human word response

Research proposal (draft)

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Gap

Large iEEG studies of free recall describe the average response to a word: high gamma rises first in visual cortex and later in prefrontal cortex (Burke et al., 2014; Kucewicz et al., 2019), and theta, alpha/beta and gamma power are induced at largely separate sites, in separate sequences ("mosaic"; 139 subjects; Marks et al., 2021). These studies average over trials and treat electrodes as samples of a region; Marks et al. (2021) also normalize each trial within its own epoch rather than against a pre-word baseline. How much of the response differs between subjects, electrodes, sessions and trials, whether its spectral components are independent beyond space, and what changes it from trial to trial have not been measured. With 362 subjects, 1009 sessions and ∼300 words per session, this dataset can answer these questions. Memory effects are out of scope (see DeHaan, 2026).

Aims

Aim 1. Where does the variability sit? Split the single-trial word response (word on vs. the electrode's usual pre-word level) into subject, electrode, session, task and trial components per region, with one random-intercept model as DeHaan (2026) used for recall. *Preliminary (40 subjects, 5 electrodes per subject and lobe; Fig. 1):* most single-trial variance is private to one electrode on one trial (alpha 61–87%, high gamma 38–79%). The next largest part is a trial's deviation shared by all of a lobe's electrodes (alpha 9–34%, high gamma 4–15%), then list-to-list drift within a session (2–6%). Task structure explains almost none of it: serial position ≤0.2%, its electrode-specific form ≤0.5%, list number, the word itself, blank length and category ≤0.2%. Stable differences are largest in occipital high gamma (subject 15%, electrode 26%, electrode × session 14%); for alpha outside occipital cortex they are near zero.

Aim 2. What is the stable part? Classify electrodes by their alpha and high-gamma response, map the types and their latency, and test whether an electrode keeps its type across sessions. *Preliminary:* outside occipital cortex 65–80% of electrodes show neither response (Fig. 2). An alpha decrease with a high-gamma increase is the most common joint type in occipital cortex (21%) but rare elsewhere (1–2%), and in occipital cortex it occurs about as often as independent responses would give (1.2×). An electrode's response pattern replicates between session halves (Fig. 4; median $r = 0.90$ in occipital high gamma).

Aim 3. Are alpha and high gamma independent beyond space, and what moves them from trial to trial? Marks et al. (2021) found the bands at separate sites. *Preliminary:* at the same electrode they are also independent trial by trial (Fig. 3; median r between -0.04 and

0.06 in every lobe). Along the list the high-gamma response shrinks while alpha rises (Fig. 5), and the high-gamma peak drops without a consistent change in latency (Fig. 6). The blank interval has small occipital effects (Fig. 7) that can be tested against recovery from adaptation (Groen et al., 2022) vs. temporal expectation. Planned: what the shared trial-to-trial component is (arousal, attention, or a common reference), single-trial latency, and the same tests per response type.

Open issues

- Trial-level artifact rejection must be added to the main pipeline (one artifact trial distorts Fig. 8); the preliminary analyses drop robust outlier trials.
- The model fit is preliminary (5 random electrodes per subject and lobe, no confidence intervals); the full fit and Wald or bootstrap intervals are next.
- The session share includes the sampling noise of each session's average; split-half reliability will separate it. Over time and frequency the split needs per-trial time–frequency power (not cached).

Figures (most to least relevant)

Status in brackets: *prelim.* = 40 random subjects; *full* = all subjects; *paper* = from the word-onset paper.

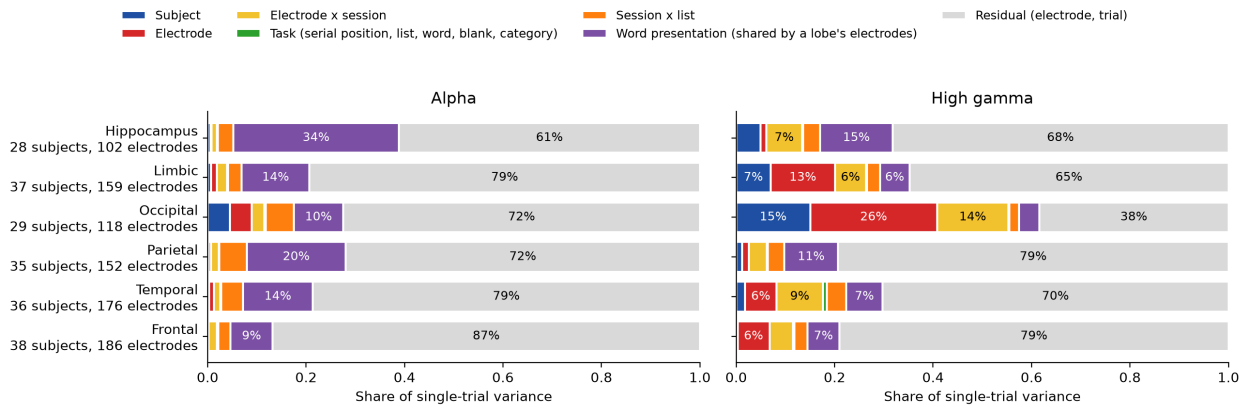


Figure 1: [prelim.] Variance components of the single-trial response, one random-intercept model per lobe and band (as DeHaan, 2026 ch. 3 for recall; up to 5 random electrodes per subject). Task: serial position, its electrode-specific form, list number, the word, blank length and category together. Word presentation: a trial's deviation shared by all of a lobe's electrodes; residual: what is left per electrode and trial.

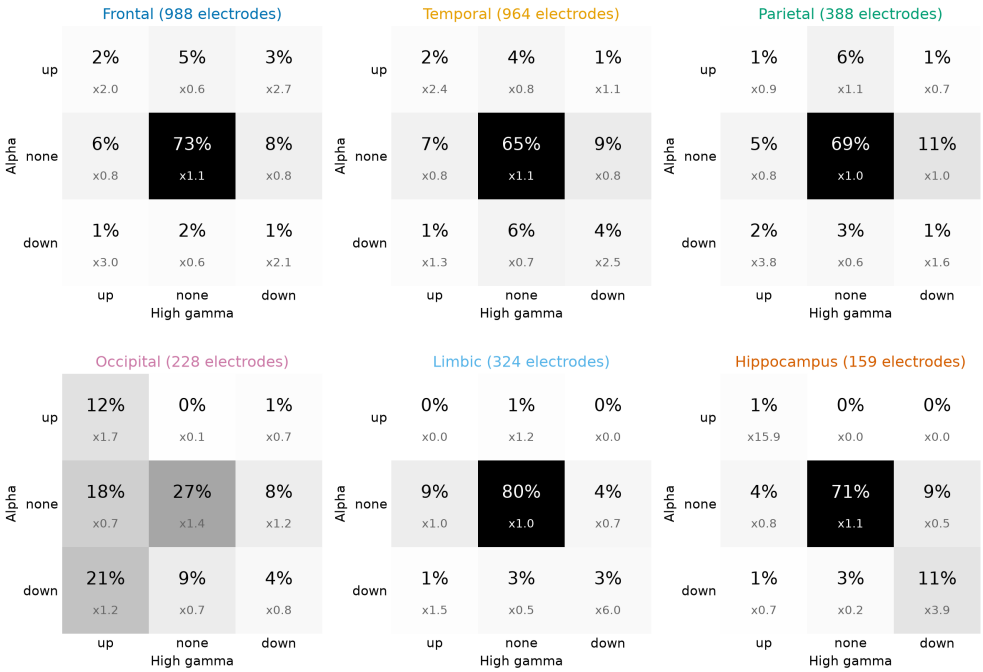


Figure 2: [prelim.] Response types: each electrode's alpha and high-gamma response classified as up, none or down (paired t over trials, word on vs. off, FDR 0.05); fraction of each lobe's electrodes (electrode count in brackets) and, below, the ratio to what independent alpha and high-gamma responses would give.

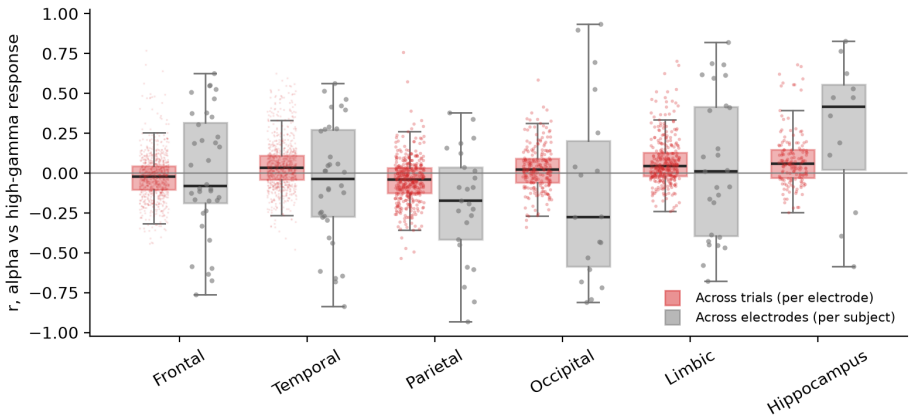


Figure 3: [prelim.] Alpha vs. high-gamma response: Pearson r across trials at each electrode (red; dots: electrodes) and across a lobe's electrodes within each subject (grey; dots: subjects).

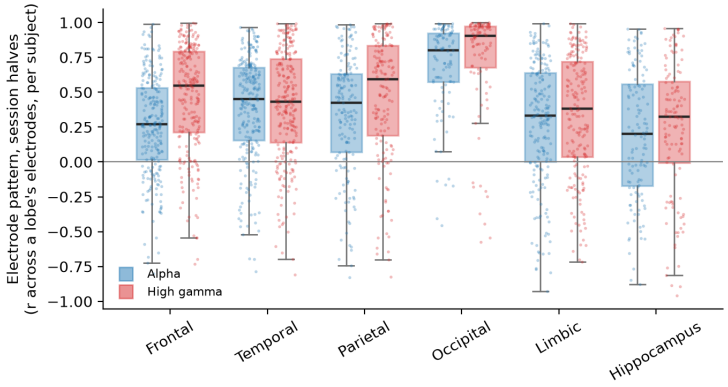


Figure 4: [full] An electrode's response pattern replicates between session halves (r across a lobe's electrodes; dots: subjects).

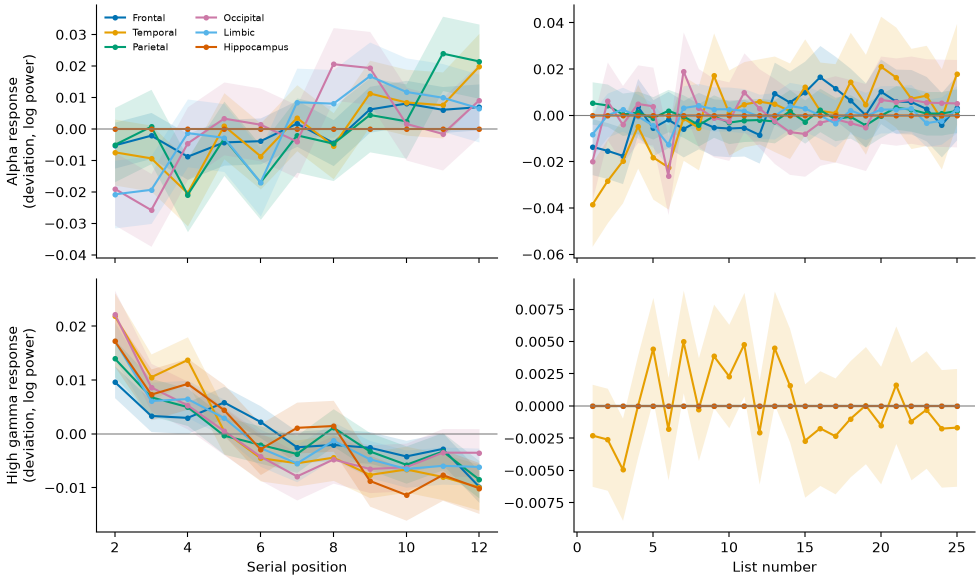


Figure 5: [prelim.] Task effects from the model of Fig. 1: estimate (± SE) per serial position and list number, per lobe.

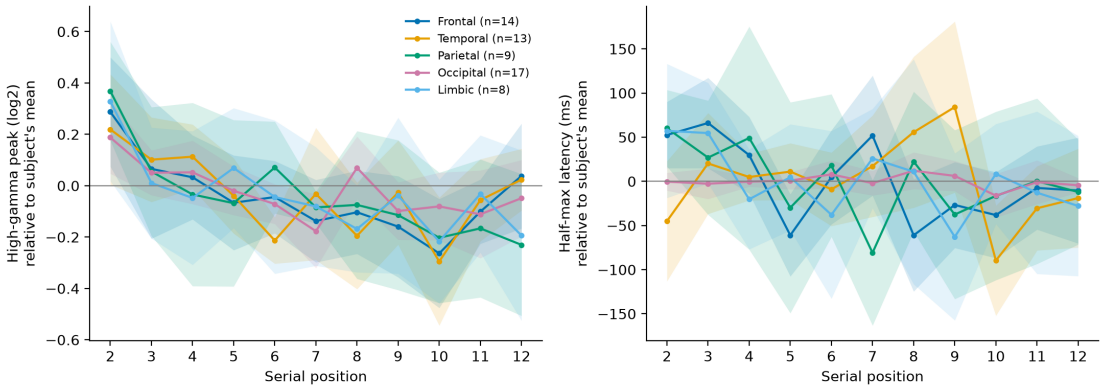


Figure 6: [prelim.] Size vs. timing across the list: high-gamma peak (left) and half-max latency (right) of responsive electrodes at each serial position, relative to each subject's mean over positions; mean and 95% CI across subjects.

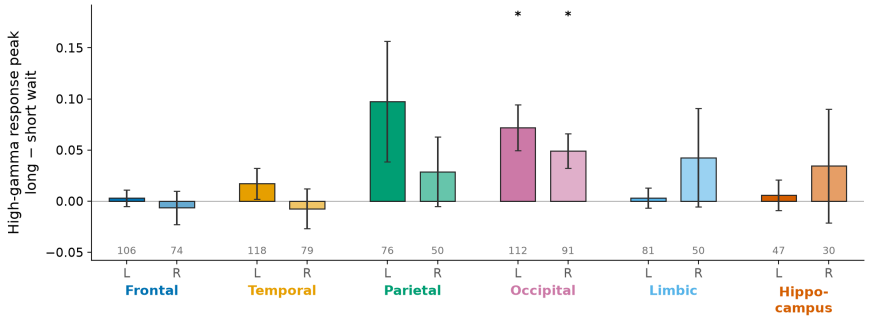


Figure 7: [paper] Occipital high gamma after long vs. short blank intervals.

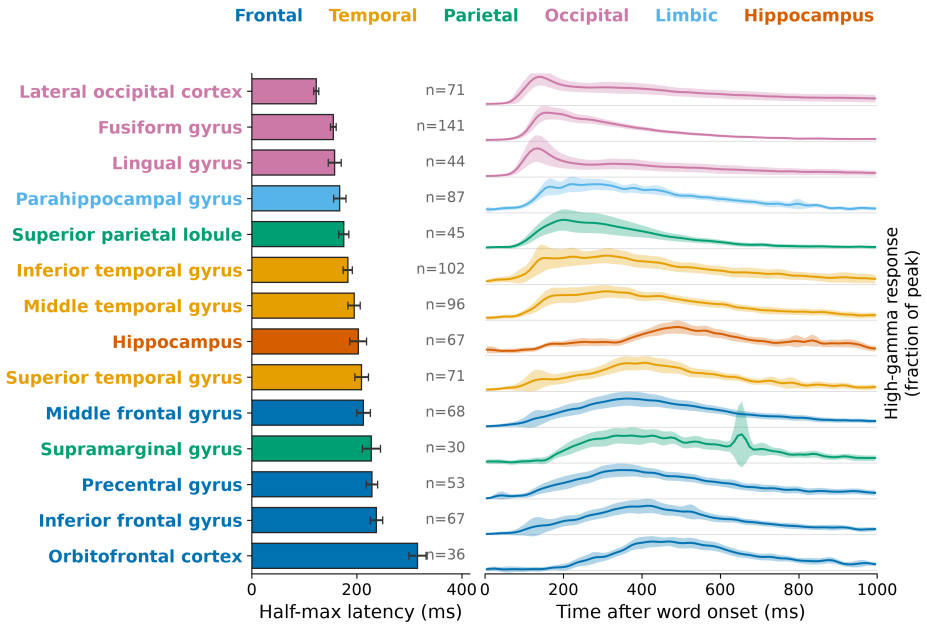


Figure 8: [paper] High-gamma latency per region (the supramarginal spike at 650 ms is one artifact trial).

References

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